

Data Analytics Technology

Datawarehouses

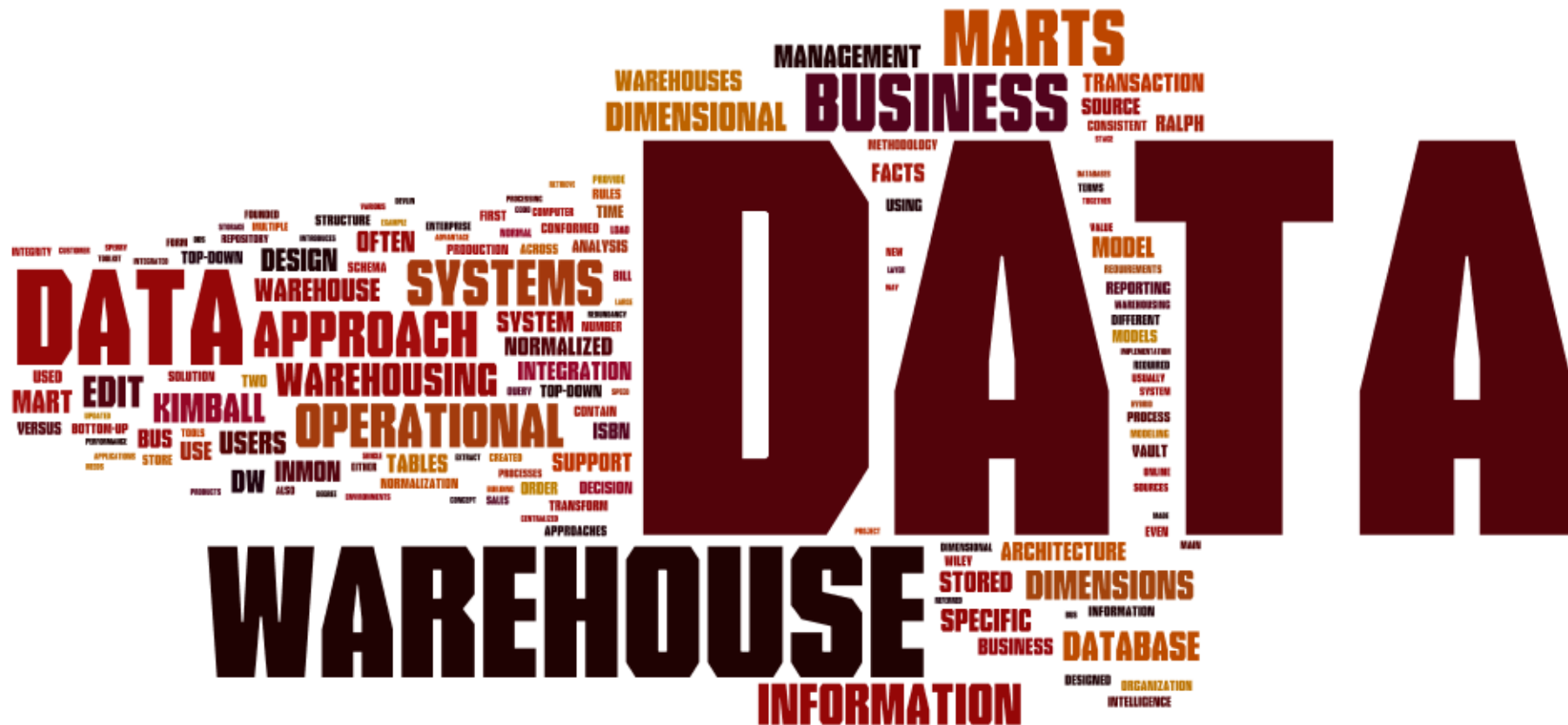
Catarina Silva

catarina@dei.uc.pt

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“Wordle” of Data Warehouses



What is a DW? (1)

- A data warehouse is a type of data management system that is designed to **enable and support business activities**, especially **analytics**.
- Data warehouses are solely intended to perform queries and analysis and often contain **large amounts of historical data**.
- The data within a data warehouse is usually derived from a **wide range of sources** such as application log files and transaction applications.

<https://www.oracle.com/pt/database/what-is-a-data-warehouse/>

What is a DW? (2)

- A data warehouse **centralizes** and **consolidates** large amounts of data from multiple sources.
- Its **analytical capabilities** allow organizations to derive valuable business insights from their data to improve **decision-making**.
- Over time, it builds a historical record that can be invaluable to data scientists and business analysts. Because of these capabilities, a data warehouse can be considered an organization's "**single source of truth**."

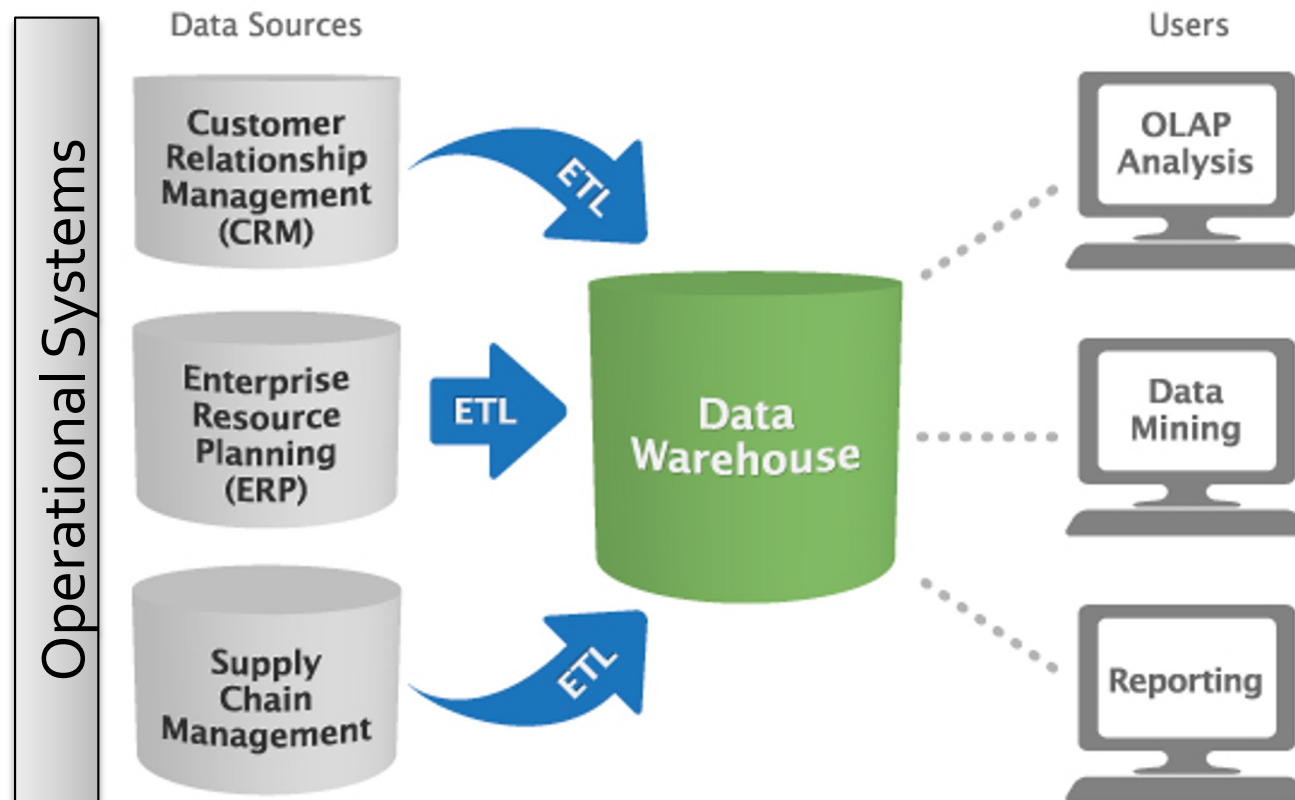
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Datawarehouse elements

- A **relational database** to store and manage data
- An extraction, transformation, and loading (**ETL**) solution for preparing the data for analysis
- Statistical analysis, reporting, and data mining capabilities
- Client analysis tools for visualizing and presenting data to users
- Other, more sophisticated analytical applications that generate actionable information by applying data science and artificial intelligence (AI) algorithms, or graph and spatial features that enable more kinds of analysis of data at scale

<https://www.oracle.com/pt/database/what-is-a-data-warehouse/>

What is a DW?



DW defining features



Subject-oriented. They can analyze data about a particular subject or functional area (such as sales).



Integrated. Data warehouses create consistency among different data types from disparate sources.



Nonvolatile. Once data is in a data warehouse, it's stable and doesn't change.



Time-variant. Data warehouse analysis looks at change over time.

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Enterprise datawarehouses

Step	Capability	Business Value
1	Transactional reporting	Provides relational information to create snapshots of business performance
2	Slice and dice, ad hoc query, BI tools	Expands capabilities for deeper insights and more robust analysis
3	Predicting future performance (data mining)	Develops visualizations and forward-looking business intelligence
4	Tactical analysis (spatial, statistics)	Offers “what-if” scenarios to inform practical decisions based on more comprehensive analysis
5	Stores many months or years of data	Stores data for only weeks or months

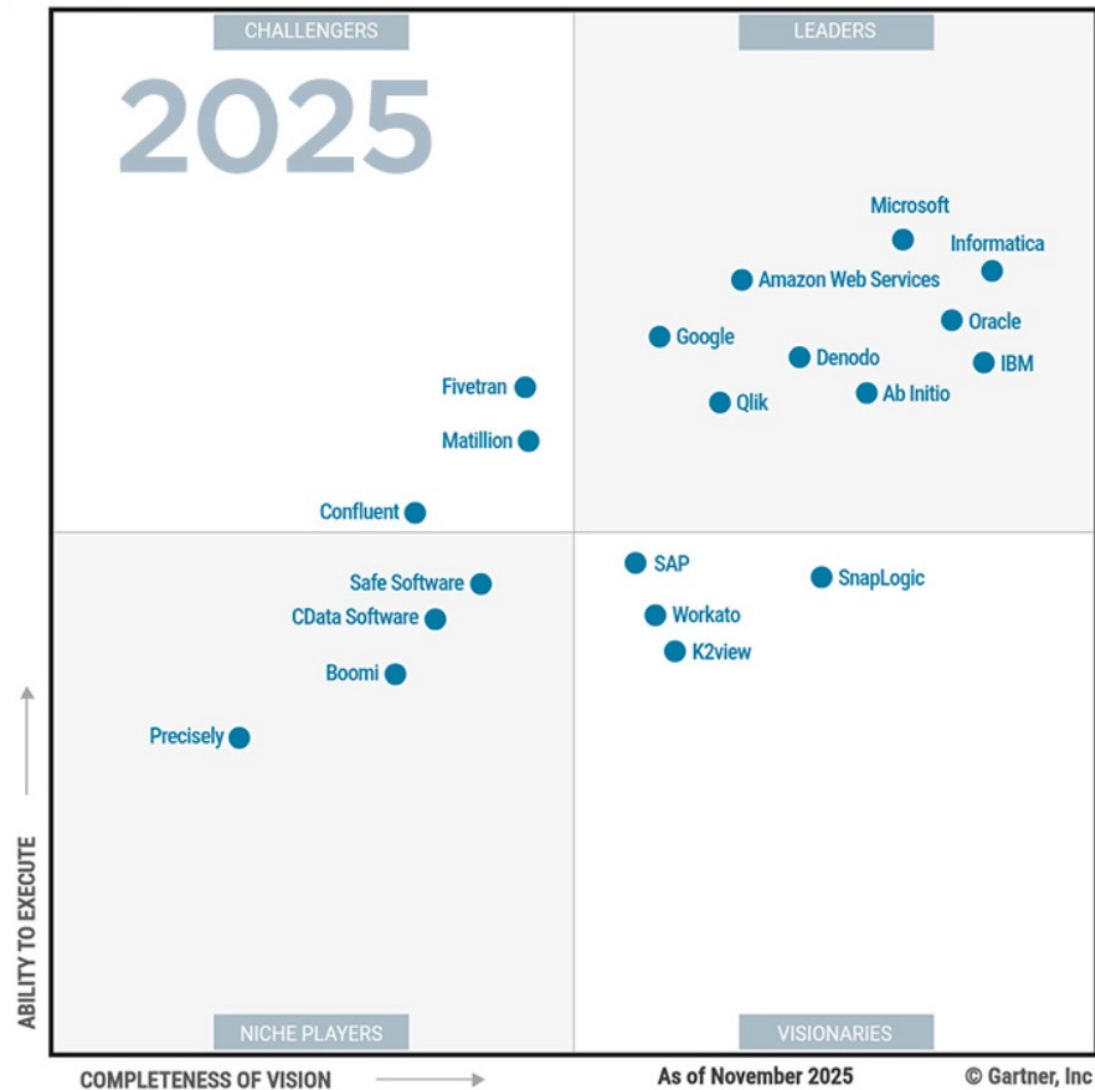
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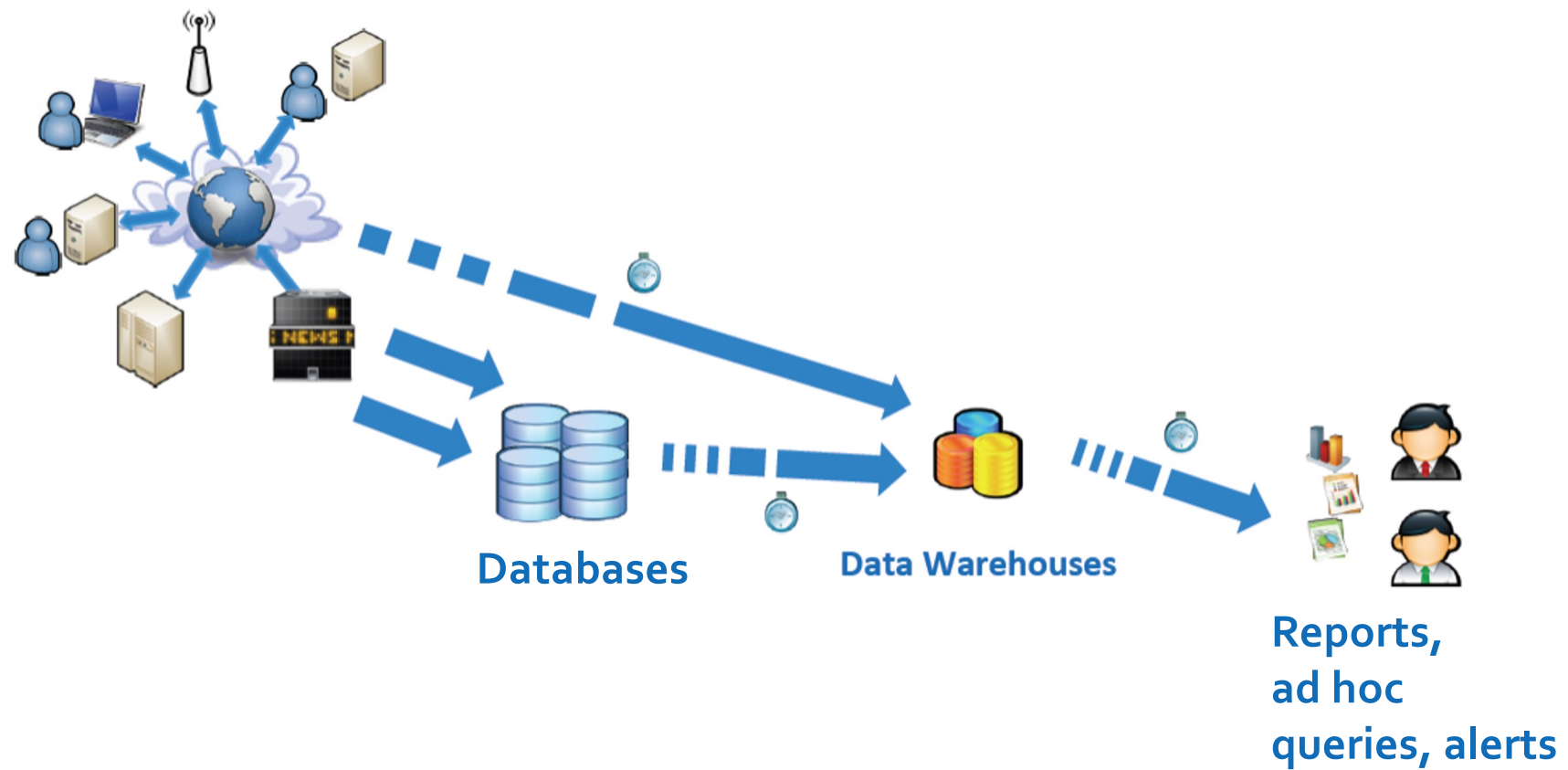
Why a DW?

- What is our best/worst selling product?
- What kind of clients are the best/worst buyers?
- What is our churn rate?
- Is a promotion being successful?
- Compare performance of different stores, seasons, years, etc...

Gartner Magic Quadrant for Data Data Integration Tools



DW in practice



OLTP vs OLAP (2)

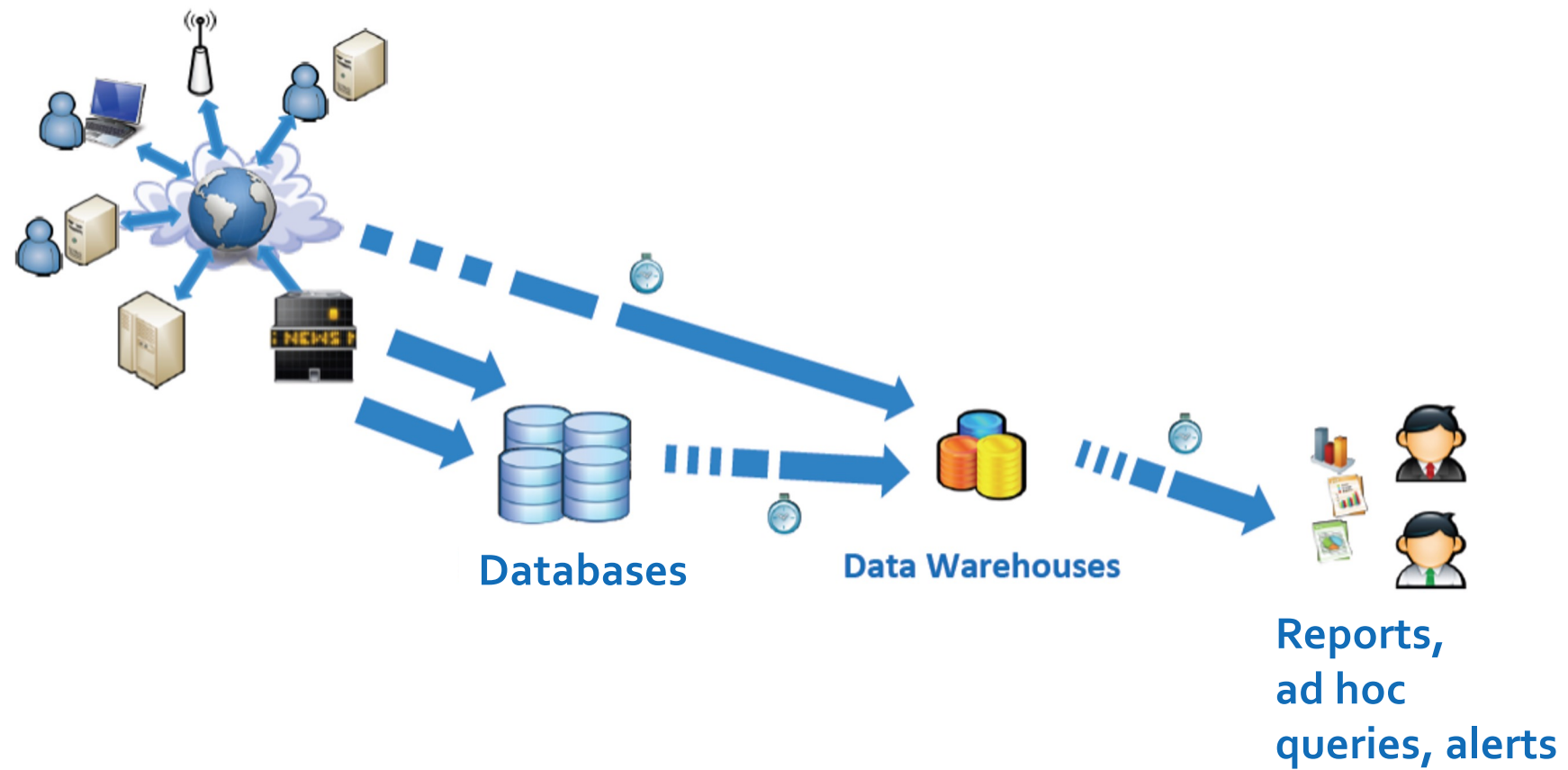
	Data Warehouse	OLTP System
Workload	Accommodates ad hoc queries and data analysis	Supports only predefined operations
Data modifications	Automatically updates on a regular basis	Updates by end users issuing individual statements
Schema design	Uses partially denormalized schemas to optimize performance	Uses fully normalized schemas to guarantee data consistency
Data scanning	Encompasses thousands to millions of rows	Accesses only a handful of records at a time
Historical data	Stores many months or years of data	Stores data for only weeks or months

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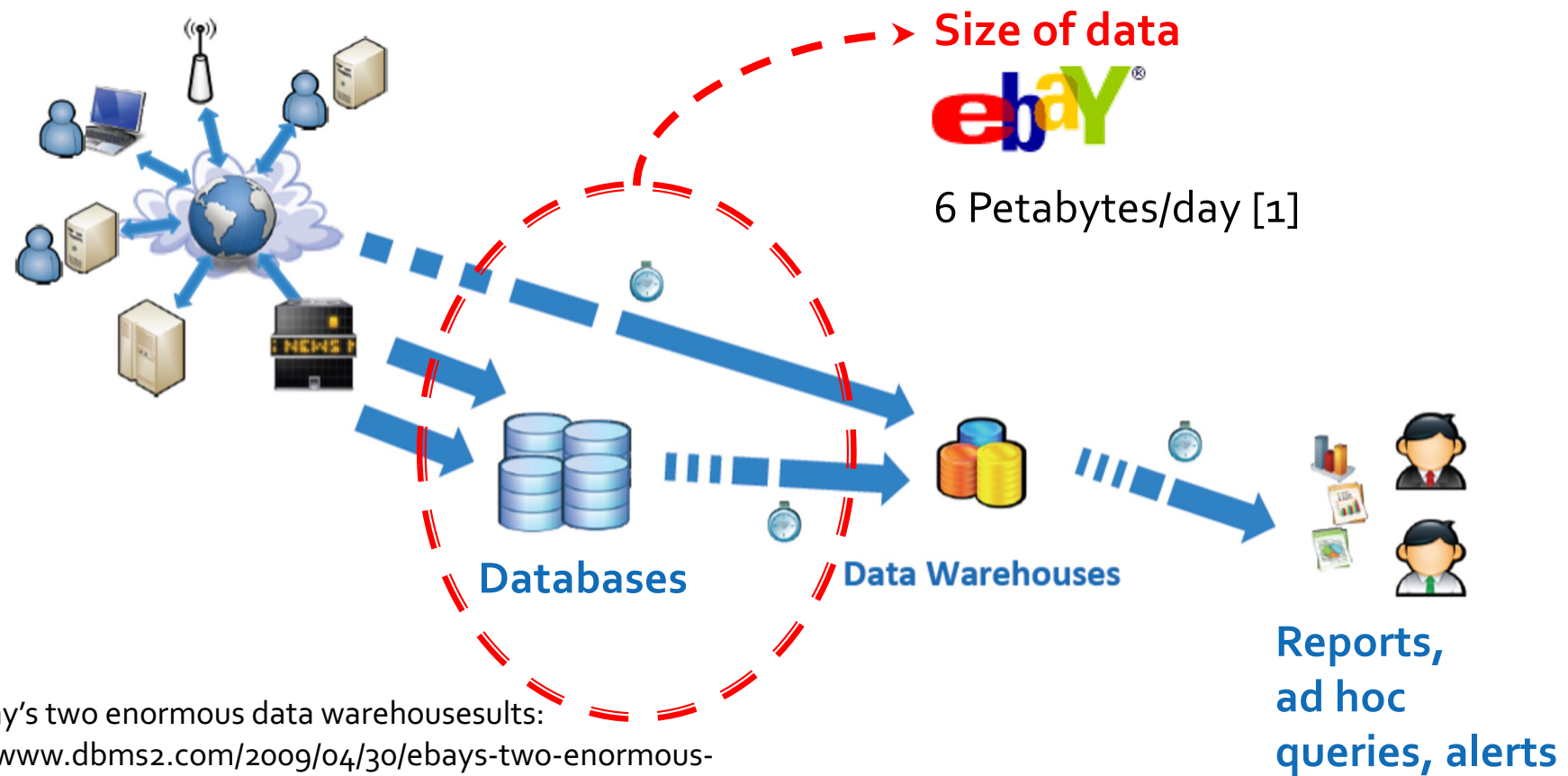
OLTP vs OLAP (3)

- OLTP – DBMS Operational
 - When it stops, business stops
 - Simultaneous readings and writings
 - Predefined transactions
 - Accesses a small number of records accessed each time
 - Data updated in real-time
 - Optimized for updates
 - Event-driven: processes generate data
- OLAP – Data Warehouse
 - No impact in operations
 - Either reading or writing
 - Ad hoc queries and reports
 - Many records in each access
 - Periodical loading of data
 - Optimized for querying
 - Data-driven: data generate answers

Datawarehouses

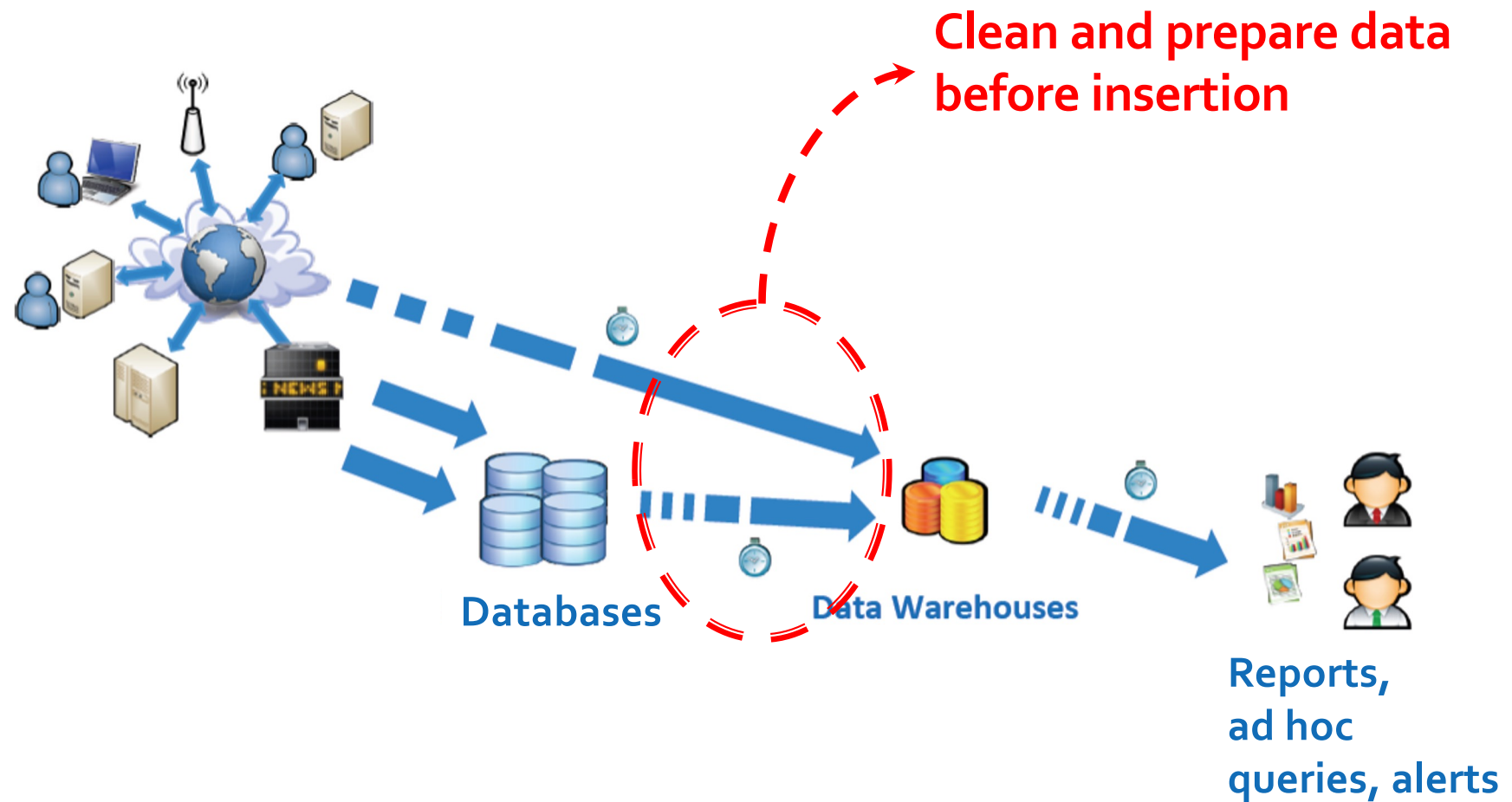


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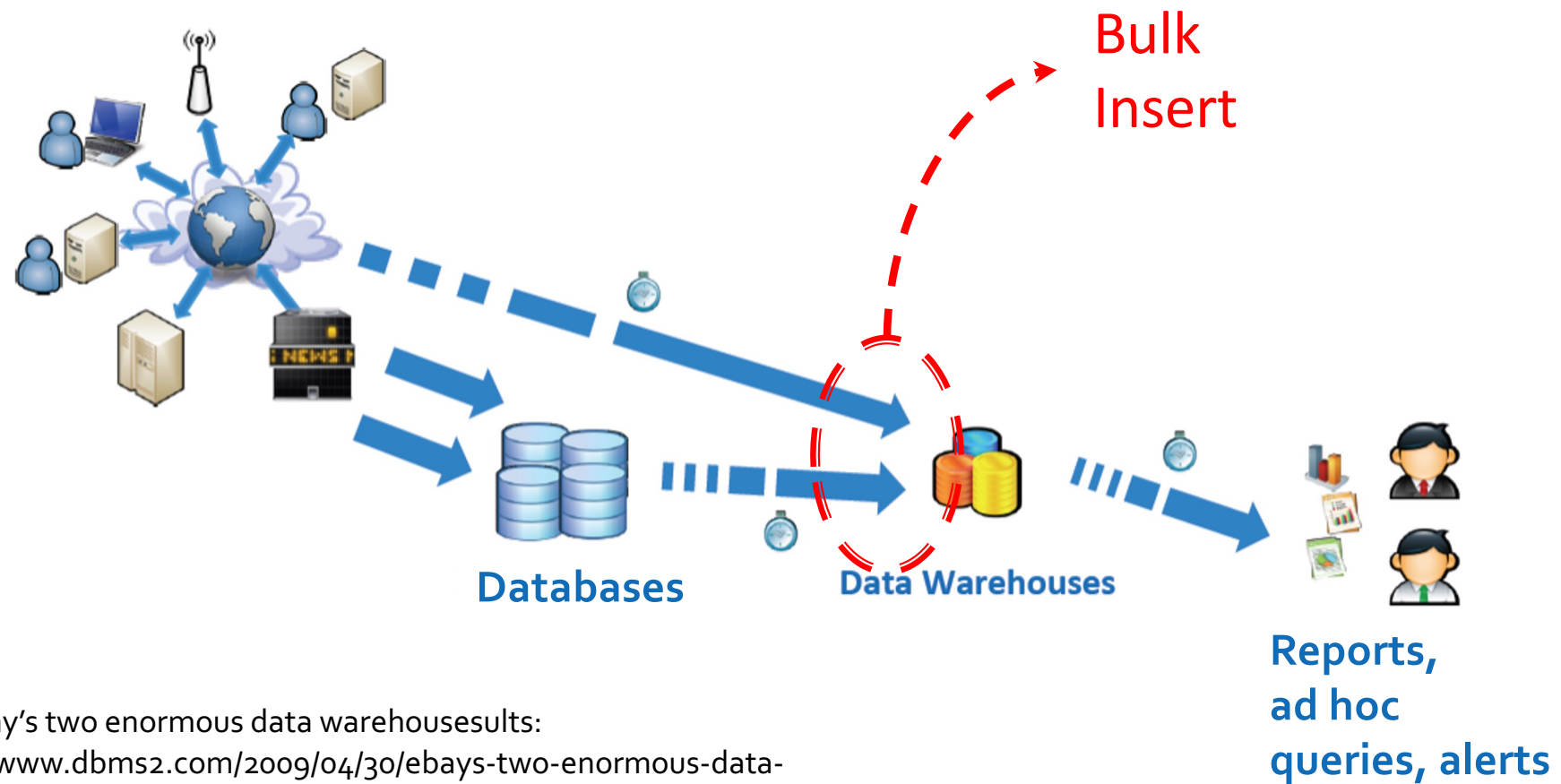


[1] eBay's two enormous data warehouses:
<http://www.dbms2.com/2009/04/30/ebays-two-enormous-data-warehouses/>

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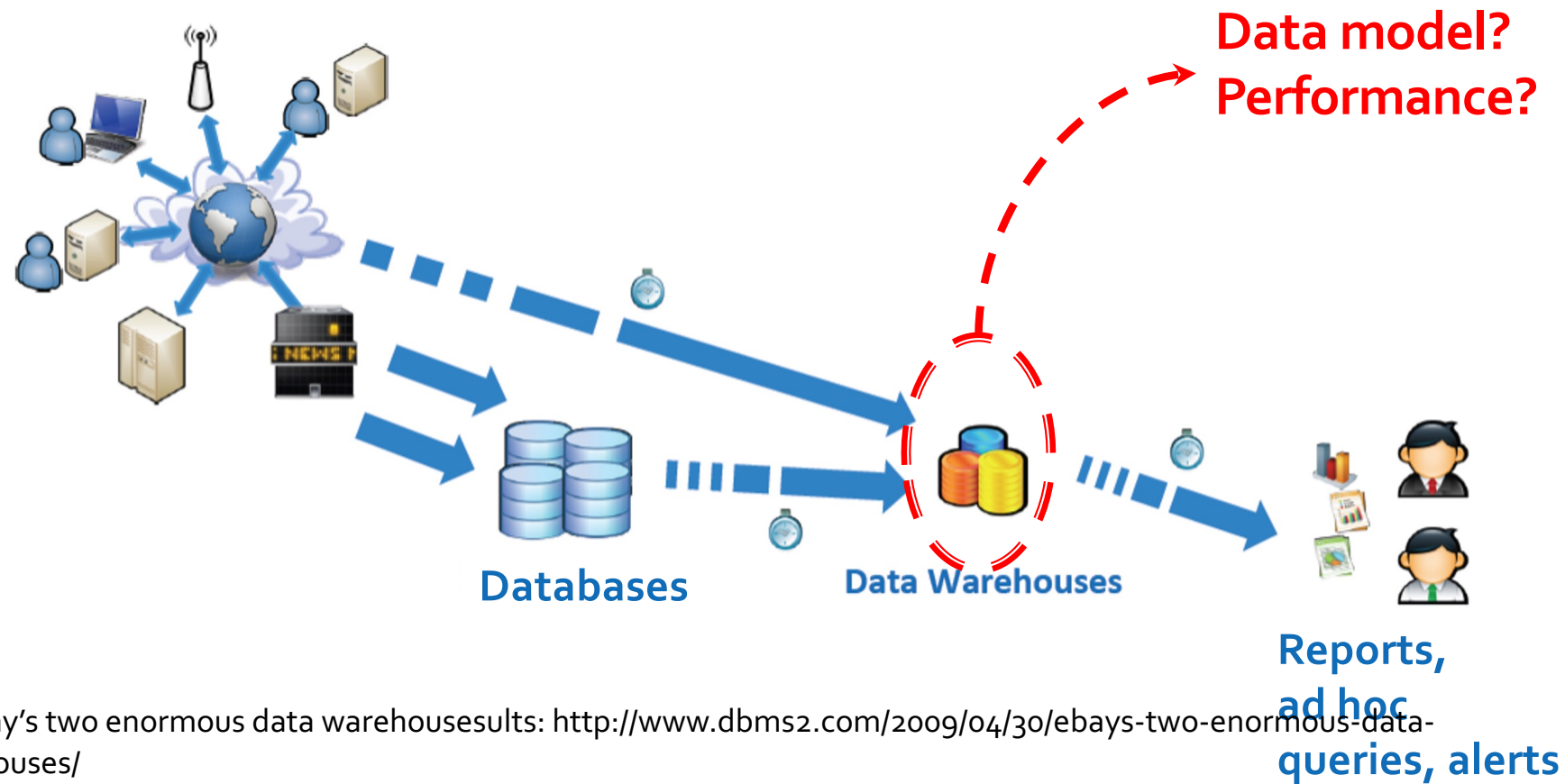


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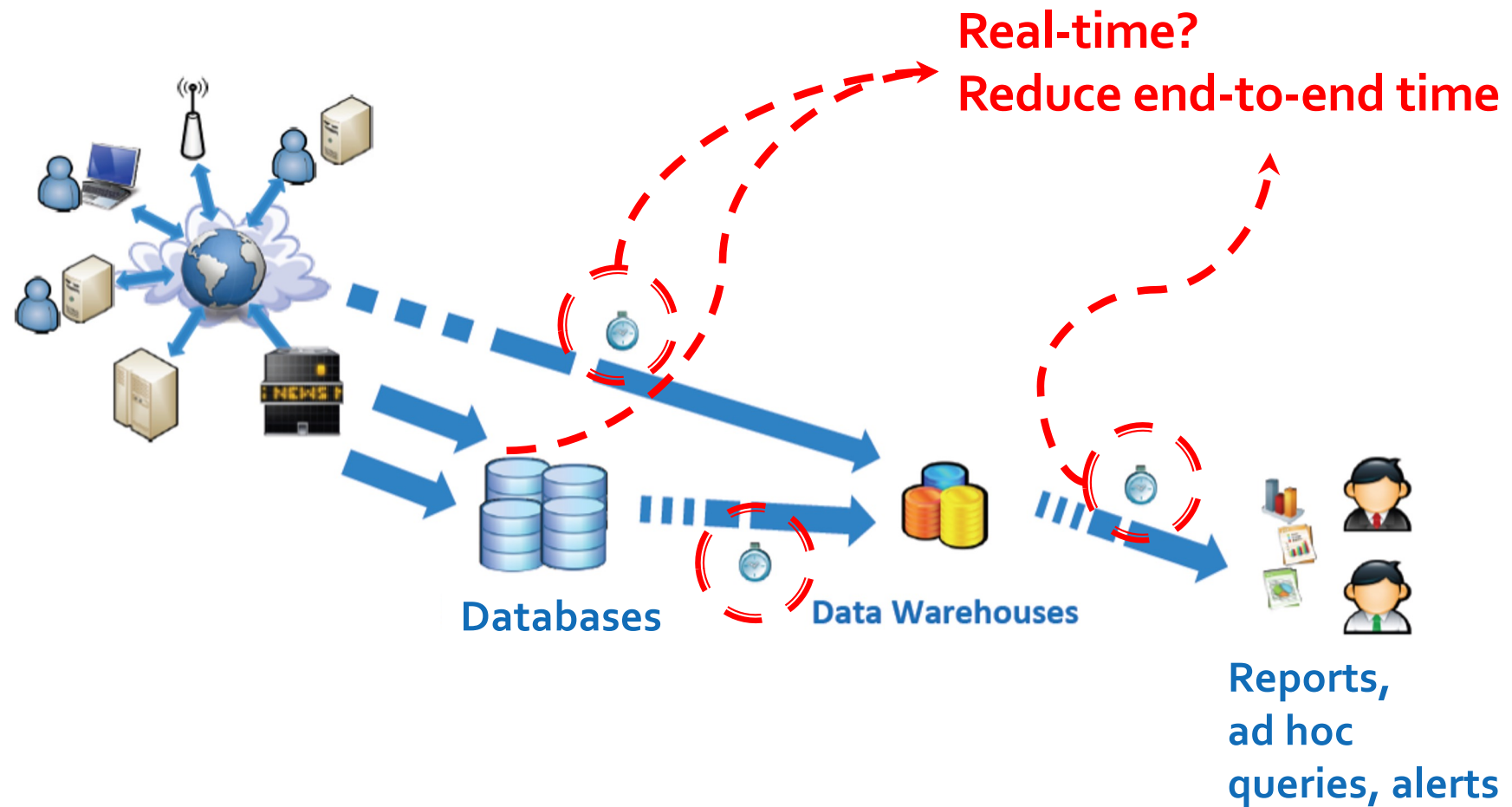


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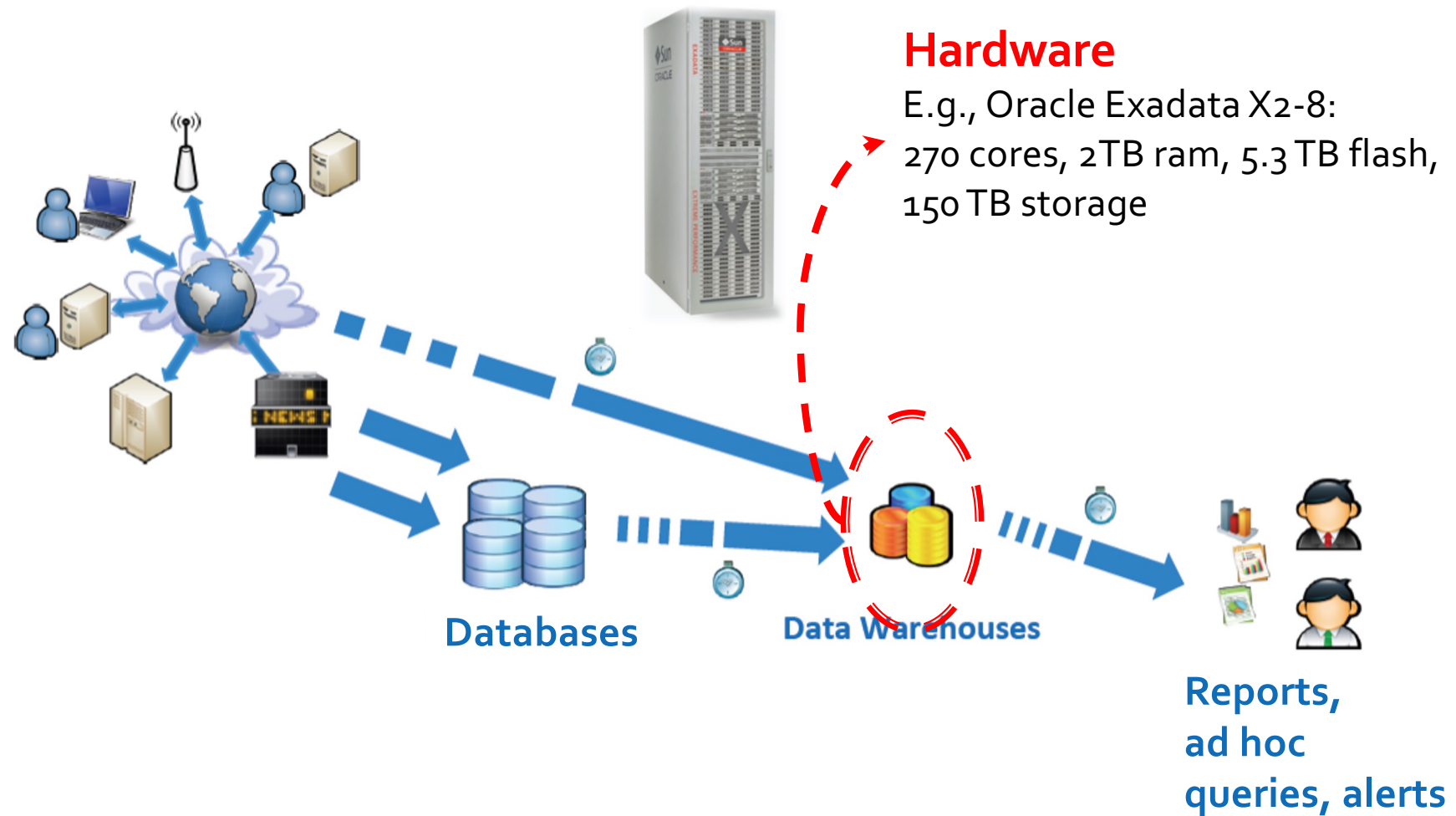
Datawarehouses



Datawarehouses



Datawarehouses



NoSQL / Big Data

- NoSQL
 - Wide Column Store / Column Families
 - Key Value / Tuple Store
 - Eventually Consistent Key Value Store
 - Document Store
 - Graph Databases



